

FPGAs and One-Hot State Assignment

- **Hardware reduction of FPGA:**
 - When designing w/ FPGAs, try to reduce the total # of **logic cells** used and the **interconnections** b/t cells instead of reducing the # of **flip-flops**.
 - To design faster logic, reduce the # of **cells** required to realize each equation.
- **One-hot state assignment:** one flip-flop per state
 - Takes more flip-flops than **encoded assignment**
 - is not an issue as most FPGA logic blocks contain 2 flip-flops.
 - **Next-state equations** for f-fs and **output equations** are often simpler than that in the **encoded method**.

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Example

- Consider the partial state graph b
- Next state equation for f-f Q_3 :

$$Q_3^+ = X_1 Q_0 Q_1' Q_2' Q_3' + X_2 Q_0' Q_1 Q_2' + X_3 Q_0' Q_1' Q_2 Q_3' + X_4 Q_0' Q_1' Q_2' Q_3$$

- Since $Q_0 = 1$ implies $Q_1 = Q_2 = Q_3 = 0$, the $Q_1' Q_2' Q_3'$ term is redundant and can be eliminated.

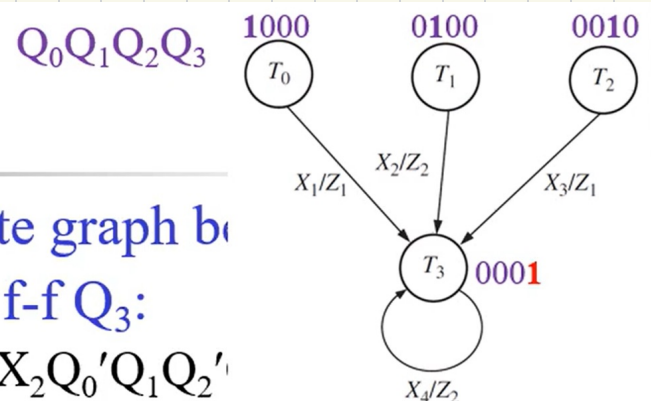
⇒ All the primed state variables can be eliminated from other terms.

$$\Rightarrow Q_3^+ = X_1 Q_0 + X_2 Q_1 + X_3 Q_2 + X_4 Q_3$$

* **Each term contains exactly one state variable.**

- Output equations of Z_1 and Z_2 :

$$Z_1 = X_1 Q_0 + X_3 Q_2, \quad Z_2 = X_2 Q_1 + X_4 Q_3$$



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System Resetting

- Resetting a system applying one-hot assignment:
 - Requires that **one flip-flop** be **set to 1** instead of resetting all flip-flops to 0.
 - If flip-flops used do **not** have a *preset* input:
 - i. Modify the one-hot assignment by replacing Q_0 w/ Q_0'

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System Resetting

- Resetting a system applying one-hot assignment:
 - Requires that **one flip-flop** be **set to 1** instead of resetting all flip-flops to 0.
 - If flip-flops used $T_0: Q_0Q_1Q_2Q_3=1000, T_1: 0100, T_2: 0010, T_3: 0001$
 - i. Modify the one-hot assignment by replacing Q_0 w/ Q_0'

$$Q_3^+ = X_1Q_0 + X_2Q_1 + X_3Q_2 + X_4Q_3$$

$$Z_1 = X_1Q_0 + X_3Q_2$$

$$Z_2 = X_2Q_1 + X_4Q_3$$

➤ E.g.:

$$T_0: Q_0Q_1Q_2Q_3 = 0000, T_1: 1100, T_2: 1010, T_3: 1001$$

$$\Rightarrow Q_3^+ = X_1Q_0' + X_2Q_1 + X_3Q_2 + X_4Q_3$$

$$Z_1 = X_1Q_0' + X_3Q_2, Z_2 = X_2Q_1 + X_4Q_3$$

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$T_0: Q_0Q_1Q_2Q_3=1000, T_1: 0100, T_2: 0010, T_3: 0001$

$$Q_3^+ = X_1Q_0 + X_2Q_1 + X_3Q_2 + X_4Q_3$$

$$Z_1 = X_1Q_0 + X_3Q_2$$

$$Z_2 = X_2Q_1 + X_4Q_3$$

ii. Add an *extra term* to the equation for the flip-flop, which should be 1 in the *starting state*.

➤ E.g.: If the system is *reset* to state 0000 after power-up (Assume that the starting state is Q_0 .)

⇒ Add the term $Q_0'Q_1'Q_2'Q_3'$ to the equation for Q_0^+ . ($Q_0^+ = Q_0'Q_1'Q_2'Q_3' + \dots$)

After the first clock the state will change from 0000 to 1000 (T_0), which is the correct starting state.

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■ *One-hot assignment:* (✓ for FPGA)

- Disadv.: More *next-state equations* are required.
- Adv: In general, both the *next-state* and *output equations* will contain fewer variables.
 - An equation with fewer variables generally requires *fewer logic cells* to realize.
 - The fewer cells are cascaded, the *shorter* the *propagation delay* and the *faster* the *operation*.